「Section 4.3

Let (X,d) be a Metric space, and let $f: X \to \mathbb{R}$ be a Continuous map.
Prove the zero set of f

$$Z(f) := \{x \in X \mid f(x) = 0\}$$

is closed.

proof. Let $y \in \overline{Z(f)}$ be given. That is, for any $r>0$,

$$B_r(y) \cap Z(f) \neq \phi.$$

Given $\varepsilon>0$, there exists $\delta>0$ s.t

$$d(t,y) < \delta \Rightarrow |f(t) - f(y)| < \varepsilon$$

For fixed $\delta>0$, Choose

$$x \in B_\delta(y) \cap Z(f)$$

Then $d(x,y) < \delta$ and $f(x) = 0$. Therefore

$$|f(x) - f(y)| = |f(y)| < \varepsilon$$

Since $\varepsilon>0$ was arbitrary, $f(y) = 0$. That is, $y \in Z(f)$.
Alternatively, $Z(f) = f^{-1}[\{0\}]$.

「Section 4.2

Let X, Y be Metric spaces. If $f: X \to Y$ is Continuous, then for any $E \subset X$,

$$f[\overline{E}] \subseteq \overline{f[E]}.$$

proof. Let $f(x) \in f[\overline{E}]$ be given. Since f is Continuous, given $\varepsilon>0$, there exists $\delta>0$ s.t

$$t \in B_\delta(x) \Rightarrow f(t) \in B_\varepsilon(f(x))$$

Meanwhile, since $x \in \overline{E}$, $B_\delta(x) \cap E \neq \phi$

Chose $u \in B_\delta(x) \cap E$. Then, $f(u) \in B_\varepsilon(f(x))$ and $f(u) \in f[E]$. That is

$$f(u) \in B_\varepsilon(f(x)) \cap f[E] \neq \phi.$$

This implies $f(x) \in \overline{f[E]}$.

「Section 4.4

Let (X, d_X) and (Y, d_Y) be a Metric space, and $f, g : X \to Y$ be a Continuous map.

If $E \subseteq X$ is a dense in X, then $f[E]$ is dense in $f[X]$.

Moreover, If $f|_E = g|_E$, then $f = g$.

Proof. Since f is Continuous, $f[X] = f[\overline{E}] \subseteq \overline{f[E]}$ is clear.

Let $x \in X = \overline{E}$ be given. If $f(x) = g(x)$, there is nothing to prove.

Suppose $f(x) \neq g(x)$. Take $\varepsilon = d(f(x), g(x)) > 0$. Since f and g are Continuous, there exist $\delta > 0$ s.t

$$t \in B_\delta(x) \Rightarrow d(f(x), f(t)),\ d(g(x), g(t)) < \frac{\varepsilon}{2}.$$

Since E is dense in X, there is $t \in B_\delta(x) \cap E$. Now,

$$d(f(x), g(x)) \leq d(f(x), f(t)) + \underbrace{d(f(t), g(t))}_{=0} + d(g(t), g(x)) < \varepsilon = d(f(x), g(x))$$

It is Contradiction.

「Thm. Let $f: X \to Y$ be a map defined on a Compact Metric space X, and any metric space Y.

If f is Continuous, then it is Continuous Uniformly.

Proof. Suppose f is continuous. Given $\varepsilon > 0$, for any $x \in X$, there exists $\delta_x > 0$ s.t

$$t \in B_{\delta_x}(x) \Rightarrow f(t) \in B_{\frac{\varepsilon}{2}}(f(x))$$

Since the set $\{B_{\frac{\delta_x}{2}}(x) \mid x \in X\}$ is open covering of X and X is Compact, there exists a finite subcover of X:

$$\{B_{\frac{\delta_i}{2}}(x_i) \mid i = 1, 2, \cdots, n\} \quad \text{where } \delta_i = \delta_{x_i}.$$

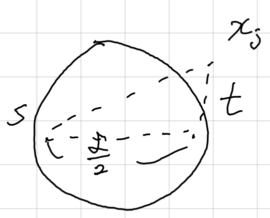
Take $\delta = \min\{\delta_1, \cdots, \delta_n\}$. Choose $s, t \in X$ s.t $d(s,t) < \frac{\delta}{2}$. (Then, $\delta_i + \delta \leq 2\delta_i, \forall i$)

Since the existence of finite subcover, for some $1 \leq j \leq n$

$$d(t, x_j) < \frac{\delta_j}{2}, \text{ which implies } d(s, x_j) \leq d(s,t) + d(t, x_j) < \frac{\delta}{2} + \frac{\delta_j}{2} \leq \delta_j$$

Finally,

$$d(f(s), f(t)) \leq d(f(s), f(x_j)) + d(f(x_j), f(t)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$



「Section 4.7. Define

$$f: \mathbb{R}^2 \to \mathbb{R} : (x,y) \mapsto \begin{cases} 0 & \text{if } (x,y)=(0,0) \\ \dfrac{xy^2}{x^2+y^4} & \text{otherwise} \end{cases}, \quad g: \mathbb{R}^2 \to \mathbb{R} : (x,y) \mapsto \begin{cases} 0 & \text{if } (x,y)=(0,0) \\ \dfrac{xy^2}{x^2+y^6} & \text{otherwise} \end{cases}$$

Show that: f is bounded on $\mathbb{R}^2$, g is not bounded in every neighborhood of $(0,0)$,
f is not continuous at $(0,0)$,
The restrictions of f, g on any straight line in $\mathbb{R}^2$ are continuous.

Proof.

g is unbounded because: for each $x \neq 0$, take $y = x^{\frac{1}{3}}$. Then,

$$g(x,y) = \frac{x \cdot x^{\frac{2}{3}}}{x^2 + x^{6 \cdot \frac{1}{3}}} = \frac{x^{\frac{5}{3}}}{2x^2} = \frac{1}{2} \cdot \frac{1}{x^{\frac{1}{3}}} \to \infty \quad \text{as} \quad x \to 0.$$

Therefore unbounded.

$$2\sqrt{xy} \leq x+y \qquad (\sqrt{x}-\sqrt{y})^2 = x+y-2\sqrt{xy}$$

f is bounded, because: $2\sqrt{x^2y^4} = 2|x| \cdot |y|^2 \leq x^2+y^4$, that is

$$\left| \frac{xy^2}{x^2+y^4} \right| \leq \frac{1}{2}.$$

(But, how can I find the bound in general setting?)

To show the discontinuity of f at $(0,0)$, take for each $x \neq 0$, $y = \sqrt{x}$. Then.

$$f(x,y) = \frac{xy^2}{x^2+y^4} = \frac{x^2}{x^2+x^2} = \frac{1}{2} \qquad \neq f(0,0).$$

Therefore, $\left(\frac{1}{n}, \frac{1}{\sqrt{n}}\right) \to (0,0)$, But $f\left(\frac{1}{n}, \frac{1}{\sqrt{n}}\right) \to \frac{1}{2}$, contradiction.

To show the last assertion, if $x = c$ for some $c \in \mathbb{R}$,

$y \mapsto \dfrac{cy^2}{c^2+y^4}$ is clearly continuous, similarly $y \mapsto \dfrac{cy^2}{c^2+y^6}$.

If $y = ax$ for some $a \in \mathbb{R}$, then

$$x \mapsto \frac{a^2x^3}{x^2+a^4x^8} \quad \text{and} \quad x \mapsto \frac{a^2x^3}{x^2+a^6x^{12}} \quad \text{are continuous.}$$

「Section 4.8 Let $f: E \to \mathbb{R}$ where $E \subseteq \mathbb{R}$ be a continuous uniformly map.

If E is bounded, then f is bounded. It is false when E is unbounded.

Proof. Let $M > 0$ be a bound of E. That is, $|a| < M$ for all $a \in E$.

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

By archimedean property, there exists $N \in \mathbb{N}$ s.t $M < \frac{\delta}{2} \cdot N$

Consider the "cover" of E:

$$C_K \overset{\text{def}}{=} \left(-\frac{\delta}{2}, \frac{\delta}{2}\right) + \frac{\delta}{2} K \qquad (K = 0, \pm 1, \cdots, \pm(N-1))$$

Then, $\forall x, y \in C_K \cap E$, $|f(x) - f(y)| < \varepsilon$ for any K. Finally,

$$\sup\{|f(x)| \mid x \in E\} \leq \max\left\{\sup_{x \in C_K \cap E} |f(x)| \;\middle|\; K = \pm 0, \cdots, \pm N-1, C_K \cap E \neq \emptyset\right\} < \infty$$

「Section 4.9. Let X, Y be metric spaces.

$f: X \to Y$ is continuous uniformly if and only if

For any $\varepsilon > 0$, there exists $\delta > 0$ s.t for all $\emptyset \neq E \subseteq X$ satisfying $\operatorname{diam} E < \delta$, $\operatorname{diam} f[E] < \varepsilon$

Proof. Suppose f is continuous uniformly. Given $\varepsilon > 0$, there exists $\delta > 0$ s.t

$$\forall x, y \in X,\ d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon/2$$

Now, for any non-empty set E satisfying $\operatorname{diam} E < \delta$,

$$d(f(x), f(y)) < \frac{\varepsilon}{2} \quad \text{for all } x, y \in E$$

This implies that $\operatorname{diam} f[E] \leq \frac{\varepsilon}{2} < \varepsilon$. The converse is trivial, because

$$d(x, y) < \delta \Rightarrow d(f(x), f(y)) = \operatorname{diam} f[\{x, y\}] < \varepsilon.$$

「Section 4.14 Let $I := [0, 1] \subseteq \mathbb{R}$

If $f: I \to I$ is continuous, then $f(x) = x$ for some $x \in I$.

Proof. Since f is continuous,

$$h: I \to \mathbb{R} : x \mapsto f(x) - x$$

is continuous. Clearly, $h(0) = f(0) - 0 \geq 0$ and $h(1) = f(1) - 1 \leq 0$.

Now, by the IVT, we obtain the result.

⌈Section 4.13. Let E be a dense set of a Metric space X. Then,
A uniformly continuous map $f: E \to \mathbb{R}$ can be extended continuously to X.

Proof.

Given $x \in X \setminus E$, Since E is dense in X, there exists a sequence $\{p_n\}$ in E s.t $p_n \to x$.
Define $\bar{f}(x) \overset{\text{def}}{=} \lim_{n\to\infty} f(p_n)$. Since $\{p_n\}$ is Cauchy sequence and f is continuous uniformly,
$\{f(p_n)\}$ is Cauchy sequence in $\mathbb{R}$. Therefore it has a limit. Moreover, if $q_n \to x$, then
given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow |\bar{f}(x) - f(p_n)| < \frac{\varepsilon}{2}$
and there exists $\delta > 0$ s.t $\forall p, q \in E$, $d(p,q) < \delta \Rightarrow |f(p) - f(q)| < \frac{\varepsilon}{2}$
and there exists $N' \geq N$ s.t

given $\varepsilon > 0$, there exists N s.t $m, n \geq N \Rightarrow d(p_m, x), d(x, q_n) < \frac{\varepsilon}{2}$
$\Rightarrow d(p_m, q_n) \leq d(p_m, x) + d(x, q_n) < \varepsilon$.

Thus $\bar{f}(x)$ is well-defined.

Define a map on X:

$$\bar{f}: X \to \mathbb{R} : x \mapsto \begin{cases} f(x) & \text{if } x \in E \\ \bar{f}(x) & \text{if } x \in X \setminus E \end{cases}$$

Given $x \in X$, and let $\varepsilon > 0$ be given, Since f is uniformly continuous on E, there is $\delta > 0$ s.t

$$\forall p, q \in E, \ d(p,q) < \delta \Rightarrow |f(p) - f(q)| < \frac{\varepsilon}{3}.$$

Let $y \in B_{\frac{\delta}{3}}(x)$ be given. Since E is dense in X, there exists $p, q \in E$ s.t

$$d(x,p) < \frac{\delta}{3} \text{ and } d(q,y) < \frac{\delta}{3}$$

$$|\bar{f}(x) - \bar{f}(y)| \leq |\bar{f}(x) - \bar{f}(p)| + |\bar{f}(p) - \bar{f}(q)| + |\bar{f}(q) - \bar{f}(y)|$$

「Section 4.13. Let E be a dense subset in a Metric space X. Then,
A uniformly Continuous map $f: E \to \mathbb{R}$ can be extended continuously to X.
Proof. Given $x \in X \setminus E$, there exists a sequence $\{p_n\}$ in E s.t $p_n \to x$.
Since f is Continuous uniformly, $\{f(p_n)\}$ is Cauchy Sequence in $\mathbb{R}$, therefore converges to $L_1 \in \mathbb{R}$.
Moreover, let $\{q_n\}$ be a sequence in E s.t $q_n \to x$. Then similarly, $\{f(q_n)\}$ converseges to L_2.
Given $\varepsilon > 0$, there exists $\delta > 0$ s.t $\forall p, q \in E$, $d(p,q) < \delta \Rightarrow |f(p) - f(q)| < \frac{\varepsilon}{3}$
Take $N_1 \in \mathbb{N}$ s.t $n \geq N_1 \Rightarrow |f(p_n) - L_1| < \frac{\varepsilon}{3}$, $N_2 \in \mathbb{N}$ s.t $n \geq N_2 \Rightarrow |f(q_n) - L_2| < \frac{\varepsilon}{3}$.
take $N_3 \in \mathbb{N}$ s.t $n \geq N_2 \Rightarrow d(p_n, x) < \frac{\delta}{2}$, take $N_4 \in \mathbb{N}$ s.t $n \geq N_3 \Rightarrow d(x, q_n) < \frac{\delta}{2}$
Now, $n \geq \max\{N_1, N_2, N_3, N_4\} \Rightarrow d(p_n, q_n) \leq d(p_n, x) + d(x, q_n) < \delta$

$$\Rightarrow |L_1 - L_2| \leq |L_1 - f(p_n)| + |f(p_n) - f(q_n)| + |f(q_n) - L_2| < \varepsilon.$$

Therefore, $\bar{f}(x) \overset{\text{def}}{=} \lim_{n\to\infty} f(p_n)$ is well-defined. Now, define an extension

$$\bar{f}: X \to \mathbb{R} : x \mapsto \begin{cases} f(x) & \text{if } x \in E \\ \bar{f}(x) & \text{if } x \in X \setminus E \end{cases}$$

To show that $\bar{f}$ is continuous, let $x \in X$ be given. Then, there exists $\{p_n\}$ in E s.t $p_n \to x$.
If $x \in E$, $f(p_n) \to f(x)$ clearly, and if $x \notin E$, $f(p_n) \to \bar{f}(x)$ by definition.
Consequently, $\bar{f}$ is Continuous.

「Section 4.11. Let X, Y be Metric spaces.
Suppose that $f: X \to Y$ is Continuous uniformly.
If $\{x_n\}$ is Cauchy sequence, then $\{f(x_n)\}$ is Cauchy sequence.
proof. Let $\{x_n\}$ be a Cauchy sequence. Given $\varepsilon > 0$, there exists $\delta > 0$ s.t

$$d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$$

Since $\{x_n\}$ is Cauchy, there exists $N \in \mathbb{N}$ s.t

$$m, n \geq N \Rightarrow d(x_m, x_n) < \delta$$

Therefore, for $m, n \geq N$, $d(f(x_m), f(x_n)) < \varepsilon$.

「If $f:\mathbb{R}\to\mathbb{R}$ is continuous uniformly, then there exist $a,b>0$ s.t

$$|f(x)|\le a\cdot|x|+b \quad \text{for all } x\in\mathbb{R}.$$

proof. There exists $\delta>0$ s.t $|x-y|\le\delta \Rightarrow |f(x)-f(y)|<1$, by uniform continuity.
Take $a=\frac{1}{\delta}$ and $b=|f(0)|+1$. Given $x\in\mathbb{R}$, there exist $N\in\mathbb{N}$ s.t $N\delta<|x|\le(N+1)\delta$.

$$\begin{aligned}|f(x)| &\le |f(x)-f(N\delta)|+|f(N\delta)-f([N-1]\cdot\delta)|+\cdots+|f(\delta)-f(0)|+|f(0)|\\ &< N+|f(0)|+1\\ &\le \frac{1}{\delta}|x|+|f(0)|+1=a|x|+b.\end{aligned}$$

gives the result.

「Section 4.10. If $f:X\to Y$ is not continuous uniformly, then
for some $\varepsilon>0$, there exist $\{p_n\},\{q_n\}$ in X satisfying $d(p_n,q_n)\to 0$ but $d(f(p_n),f(q_n))>\varepsilon$.
Proof. Suppose that given $\varepsilon>0$, if $\{p_n\},\{q_n\}$ in X satisfies $d(p_n,q_n)\to 0$ then $d(f(p_n),f(q_n))<\varepsilon$.

「Section 4.11. Let X,Y be metric spaces.
Suppose that $f:X\to Y$ is continuous uniformly.
If $\{x_n\}$ is Cauchy sequence, then $\{f(x_n)\}$ is Cauchy sequence.
proof. Let $\{x_n\}$ be a Cauchy sequence. Given $\varepsilon>0$, there exists $\delta>0$ s.t

$$d(x,y)<\delta \Rightarrow d(f(x),f(y))<\varepsilon$$

Since $\{x_n\}$ is Cauchy, there exists $N\in\mathbb{N}$ s.t

$$m,n\ge N \Rightarrow d(x_m,x_n)<\delta$$

Therefore, for $m,n\ge N$, $d(f(x_m),f(x_n))<\varepsilon$.

「Section 4.15. Continuous open real-valued map on $\mathbb{R}$ is strictly monotonic.
Proof. Let $f: \mathbb{R} \to \mathbb{R}$ be a Continuous Open map. If f is not monotonic, then For some $x<y$, there exists $t \in (x,y)$ s.t

$$f(t) = \max_{s\in[x,y]} f(s) \quad \text{or} \quad f(t) = \min_{s\in[x,y]} f(s)$$

Therefore
$$f[(x,y)] = (\quad , f(t)] \quad \text{or} \quad [f(t), \quad),$$

which is Contradiction.
If for some open interval (a,b), $f|_{(a,b)}$ is constant, then $f[(a,b)] = \{f(\frac{a+b}{2})\}$ singleton, it is Contradiction to f is open map. Thus f is Not constant on any open interval.

「Section 4.18. Define the Thomae function as:

$$f : \mathbb{R} \to \mathbb{R} ; x \mapsto \begin{cases} 0 & x \notin \mathbb{Q} \text{ or } x = 0 \\ \frac{1}{n} & x \in \mathbb{Q} \setminus \{0\},\ x = \frac{m}{n} \text{ where } \gcd(n,m) = 1. \end{cases}$$

Then, f is continuous at every irrationals, and
f has a simple-discontinuity at every rationals.

Proof. Let $x \in \mathbb{R} \setminus \mathbb{Q}$ be given. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $1 < N \cdot \varepsilon$.
observe that $S = f^{-1}[\{\frac{1}{2}, \frac{1}{3}, \cdots, \frac{1}{N}\}]$ is a finite set, because

for any $n \in \mathbb{N}$, $f(x) = \frac{1}{n}$ iff and only if $x = \frac{1}{n}$ or $\frac{2}{n}$ or $\cdots$ or $\frac{n-1}{n}$.

Take $\delta = \min\{|x-a| \mid a \in S\}$. Then, this $\delta > 0$ because $x \notin S$ and S is finite.

Now, $|x-t| < \delta \Rightarrow |f(x) - f(t)| = |f(t)| < \frac{1}{N} < \varepsilon$, therefore continuous.

But, unfortunately, at rationals, f is <u>not</u> continuous, because:
Let $x \in \mathbb{Q} \setminus \{0\}$ be given. Denote $x = \frac{m}{n}$ as irreducible form. Take $\varepsilon = \frac{1}{n}$.
For any $\delta > 0$, $B_\delta(x)$ must contains an irrational element. That is,

for some $t \in B_\delta(x)$, $|f(x) - f(t)| = \frac{1}{n} \nless \varepsilon$

Thus, f is not continuous at every rationals, except $x = 0$.
But the side-limits exist, because for any sequence $\{x_n\}$ in (x, ∞) s.t $x_n \to x$,

$\lim_{n \to \infty} f(x_n) = 0$ because: for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $\frac{1}{N} < \varepsilon$ and

except finite terms of $f(x_n)$, $f(x) \leq \frac{1}{N}$. Thus $f(x+)$ exists, similarly $f(x-)$ exists.

Note: for any $n \in \mathbb{N}$, a set

$$A_n \overset{\text{def}}{=} \{x \in \mathbb{R} \mid f(x) \geq \tfrac{1}{n}\} = f^{-1}[\{\tfrac{1}{2}, \cdots, \tfrac{1}{n}\}]$$

is finite set, because $f(x) \geq \frac{1}{n} \iff x = \frac{1}{1}$ or $\frac{1}{2}$ or $\frac{1}{3}$ or $\frac{2}{3}$ or $\cdots$ or $\frac{n-1}{n}$

「Section 4.19.

If $f: \mathbb{R} \to \mathbb{R}$ satisfies Intermidiate Value Property and for any rational $r \in \mathbb{Q}$, $f^{-1}[\{r\}]$ is closed, then f is Continuous. (IVT property: If $f(a) < c < f(b)$, then $\exists x \in (a,b)$ s.t $c = f(x)$)

Proof. Suppose that f is not Continuous map. Then, there exists a sequence $\{x_n\}$ which converges to x_0 but $f(x_n)$ not converges to $f(x_0)$. That is, there exists $\varepsilon > 0$ s.t

there exist infinitely many $n_i \in \mathbb{N}$ satisfying $|f(x_n) - f(x_0)| \geq \varepsilon$. $\left(\Leftrightarrow \begin{array}{c} f(x_0) \leq f(x_n) - \varepsilon < f(x_n) \\ \text{or} \\ f(x_0) \geq f(x_n) + \varepsilon > f(x_n) \end{array} \right)$

By the density of rationals, there exist two rationals $r^- \in (f(x_0) - \varepsilon, f(x_0))$ and $r^+ \in (f(x_0), f(x_0) + \varepsilon)$.

if $f(x_0) < f(x_{n_i})$, then $f(x_0) < r^+ < f(x_{n_i})$ and

if $f(x_{n_i}) < f(x_0)$, then $f(x_{n_i}) < r^- < f(x_0)$.

By the IVP, for each n_i, there exists $t_i \in (x_0, x_{n_i}) \cup (x_{n_i}, x_0)$ s.t $f(t_i) = r^+$ or r^-.

Since $x_{n_i} \to x_0$ as $i \to \infty$, $t_i \to x_0$ as $i \to \infty$.

For each $i \in \mathbb{N}$, $t_i \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}]$, closed set, therefore the limit point x_0 satisfies

$$x_0 \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}].$$

That is, $f(x_0) = r^+$ or r^-, it is Contradiction.

⌈Section 4. 20. For non-empty subset $E \subset X$ of a Metric space X, define

$$\rho_E : X \to \mathbb{R} : x \mapsto \inf_{z \in E} d(x, z)$$

a). $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.

b). ρ_E is Continuous uniformly.

Proof. $x \in \overline{E} \iff \forall \varepsilon > 0,\ B_\varepsilon(x) \cap E \neq \phi$

$\iff \forall \varepsilon > 0$, there exists $y \in E$ s.t $d(x, y) < \varepsilon$

By definition, $\rho_E(x)$ must be smaller than $d(x,y)$ for all $y \in E$, thus smaller than ε. Therefore $\rho_E(x) = 0$. To show the uniform Continuity, observe that: for any $x, y \in E$,

$$\rho_E(x) \leq d(x, z) \leq d(x,y) + d(y, z) \text{ for all } z \in E$$

which implies that $\rho_E(x) \leq d(x,y) + \rho_E(y)$. Now,

$$|\rho_E(x) - \rho_E(y)| \leq d(x,y) \text{ for all } x, y \in E.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$. Then,

$$d(x,y) < \delta = \varepsilon \Rightarrow |\rho_E(x) - \rho_E(y)| \leq d(x,y) < \varepsilon.$$

⌈21. If K, F are disjoint subset of a Metric space X, and K is compact and F is closed, then there exists $\delta > 0$ s.t for all $x \in K$, $y \in F$, $d(x,y) \geq \delta$.

Proof. Since K and F are closed and disjoint, $\rho_F(x) > 0$ for all $x \in K (= \overline{K} \subseteq \overline{F}^c = F^c)$

And, since ρ_F is Continuous, $\rho_F[K]$ has minimum in K. Therefore, $\inf \rho_E[K] > 0$.

Counterexample if K is closed but not Compact:

In $\mathbb{R}^2$, let $K = \{(x,y) \in \mathbb{R}^2 \mid y \geq e^x\}$ and $F = \{(x,y) \in \mathbb{R}^2 \mid y \leq 0\}$

「Let $A \subseteq \mathbb{R}^d$ be a non-empty. Define

$$d(\cdot, A) : \mathbb{R}^d \to \mathbb{R} : x \mapsto \inf_{y \in A} \|x-y\|$$

Prove

$$\overline{A} = \bigcap_{n=1}^{\infty} \{x \in \mathbb{R}^d \mid d(x, A) < \tfrac{1}{n}\}$$

It is still true when $\mathbb{R}^d$ be replaced to general Metric Space?

Proof. Let $x \in \overline{A}$ be given. Then, for all $n \in \mathbb{N}$,

$$A \cap B_{\frac{1}{n}}(x) \neq \phi.$$

That is, for all $n \in \mathbb{N}$, there exists $a \in A$ s.t $\|x-a\| < \frac{1}{n}$, this implies

$$d(x, A) \leq d(x, a) < \frac{1}{n}$$

Thus $x \in \overline{A}$. Conversely, let $x \in \bigcap_{n=1}^{\infty} \{x \in \mathbb{R}^d \mid d(x, A) < \frac{1}{n}\}$.

Let $\varepsilon > 0$ be given. Take $n \in \mathbb{N}$ such that $1 < \varepsilon n$
Since for all $n \in \mathbb{N}$, $d(x, A) < \frac{1}{n}$, there exist $a \in A$ s.t

$$d(x, A) \leq d(x, a) < \frac{1}{n}$$

That is, for fixed $\varepsilon > 0$, we can choose $n \in \mathbb{N}$ s.t $\frac{1}{n} < \varepsilon$ and we can find $a \in A$ s.t

$$a \in A \cap B_{\frac{1}{n}}(x) \subseteq A \cap B_{\varepsilon}(x) \neq \phi.$$

⌈22. Let $A, B \subseteq X$ be non-empty disjoint closed sets in Metric space X.
Show that a map
$$f: X \to \mathbb{R}; x \mapsto \frac{\rho_A(x)}{\rho_A(x)+\rho_B(x)}$$
is continuous $f(x)=0$ iff $x \in A$ and $f(x)=1$ iff $x \in B$.

Proof Since there is no element in X s.t
$$\rho_A(x)+\rho_B(x)=0$$
because it is equivalent to $\rho_A(x)=\rho_B(x)=0$, but it implies that
$$x \in \overline{A} \cap \overline{B} = A \cap B = \emptyset$$
by assumption, which is contradiction. Therefore this f is well-defined.
The continuity is clear. And, observe that
$$\frac{\rho_A(x)}{\rho_A(x)+\rho_B(x)} = 0 \text{ iff } \rho_A(x)=0 \text{ iff } x \in A = \overline{A}.$$
and
$$\frac{\rho_A(x)}{\rho_A(x)+\rho_B(x)} = 1 \text{ iff } \rho_B(x)=0 \text{ iff } x \in B = \overline{B}.$$

⌈Every closed subset $A \subseteq X$ is $Z(f)$ for some continuous map $f: X \to \mathbb{R}$.
Proof. If $A=X$, then put $f=0$. Suppose that $A \subsetneq X$.
Then, there exists non-empty closed set $B \subset X$ s.t $A \cap B = \emptyset$. (For example, $B=\{b\}$, $b \notin A$)
Now, $f(x)=\dfrac{\rho_A(x)}{\rho_A(x)+\rho_B(x)}$, then $Z(f)=A$.
Moreover, observe that
$$A = f^{-1}[\{0\}] \subseteq f^{-1}[[0,\tfrac{1}{2})] \text{ and } B = f^{-1}[\{1\}] \subseteq f^{-1}[(\tfrac{1}{2},1]]$$
implies that A and B are covered by disjoint open sets.

「Anothe Question

1. If f, g are continuous uniformly, then $f+g$, fg is?